

RESEARCH ARTICLE

Nanoporous Carbon Coating of Separator Boosts Rate Capability of Cathodes in Lithium-Ion Batteries

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ABSTRACT

High rate capability of cathodes in lithium-ion batteries (LIBs) is essential for fast rechargeability of electric vehicles. Herein, we report that coating the cathode side of the separator with a layer of nanoporous carbon significantly improves the rate capability of $\text{LiNi}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}\text{O}_2$ (NMC) and LiFePO_4 (LFP) cathode materials, where NMC and LFP deliver 121 mAhg^{-1} and 125 mAhg^{-1} at 2 Ag^{-1} ($\sim 12 \text{ C}$), respectively. To elucidate the underlying mechanism, we investigate the interfacial behavior and charge transfer kinetics using the distribution of relaxation times (DRT) from electrochemical impedance spectroscopy. Our DRT results show an increased relaxation time for mass transport and indicate a more populated charge storage in the diffuse layer of the electrical double layer (EDL). Simulations on ion transport behavior using a size-modified Poisson–Nernst–Planck equation reveal that the nanopore-confined EDLs produce an exclusion effect that increases Li-ion transport resistance in over-confined pores. These results support the experimentally observed increase in performance for coatings with expanded pore sizes and further indicate that the carbon layer enhances both (de)lithiation processes by promoting a more uniform and consistent gradient of Li-ions in the EDL.

1 | Introduction

With the rapid market expansion of electric vehicles (EV), demand for fast-charging Li-ion batteries (LIBs) has increased immensely [1–3]. While electrode materials predominantly dictate the battery rate performance, the electrode-electrolyte interfacial properties, often defined by separators, also play a vital role in regulating ion transport [4, 5]. Prior studies on separators have sought to improve wettability, mechanical strength, and ionic conductivity by inorganic fillers or coatings [3, 6–10]. In addition, carbon coatings have been investigated to counter the shuttle effect by retaining solvated active mass species via absorption [11–16]. Beyond the adsorption effects, carbon coating has also been reported to benefit the cathode's rate performance

in LIBs, although the underlying mechanism of the enhancement remains elusive [17].

This study shows that applying a nanoporous carbon coating to the cathode-facing side of a commercial polypropylene (PP) separator improves rate performance of both $\text{LiNi}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}\text{O}_2$ (NMC622, referred to hereafter as NMC) and LiFePO_4 (LFP) cathode materials. In this system, the porous carbon layer establishes electrical contact with the cathode surface by sharing the cathode's operation potential, which exceeds the potential of zero charge (PZC, 3.2 V vs. Li^+/Li) of the coated carbon layer. As a result, during battery operation, the nanoporous carbon layer remains positively polarized, necessitating charge balance through anion enrichment in the pores, thereby forming an

electrical double layer (EDL) [18–22]. It is hypothesized that such EDL facilitates Li^+ transport near the cathode-electrolyte interface, thereby improving rate performance of both cathode materials.

How an extended EDL on the electrode surface affects ion transport to the electrode remains an open question. Conventional modeling approaches struggle to capture the EDL structure and its influence on ion transport in nanopores, especially in systems where the pore size is of a similar scale to EDL thickness. The Gouy-Chapman-Stern model describes the EDL as a combination of a compact Stern-layer and a diffuse layer; however, most porous-electrode models implicitly assume that the EDL is much thinner than the pore width—a condition that does not hold for sub-10-nm pores [18–24]. Recent advances in modeling nanopore electrostatics provide a more accurate description by explicitly accounting for overlapping diffuse layers and their effects on local ion distributions and transport [25].

Herein, the reaction kinetics and interfacial processes are investigated using electrochemical impedance spectroscopy (EIS) combined with distribution of relaxation times (DRT) analysis of systems with and without the porous carbon coating on the separator [26]. To reveal the impacts of pore sizes of porous carbon on the ion transport, a pore model system is designed using a size-modified Poisson-Nernst-Planck relation to simulate Li-ion transport through the carbon nanopores [27, 28]. This framework allows us to quantitatively assess how nanoconfined EDLs modulate Li-ion flux across a range of pore widths under steady-state conditions.

2 | Results and Discussion

2.1 | Effects of KB-Coating on Electrochemical Performance

Galvanostatic charge-discharge (GCD) cycling was used to assess the capacity and rate performance of the electrodes employing either a plain PP separator or a PP separator coated on the cathode-facing side with a nanoporous carbon layer (Ketjenblack, KB). The carbon-coated separators had an areal carbon loading of approximately 0.25 mg cm^{-2} . For 19 mm separator disks, the average coating mass was 0.73 mg, corresponding to an average coating thickness of 9 μm , which was independently confirmed by cross-sectional SEM to be approximately 10 μm (Figure S1). As shown in Figure 1a,d, the KB-coated PP separator improves the capacity for NMC at high current rates above 500 mA g^{-1} and enhances the capacity of LFP across all testing rates. For NMC, the KB-PP does not alter the shape of the GCD profiles (Figure 1b,c). In contrast, for LFP, the KB-PP substantially reduces the overpotentials in both charge and discharge profiles in addition to promoting the capacity (Figure 1e,f), along with a more defined elbow at the end of both the charge and discharge steps that is associated with a more rapid concentration polarization due to higher utilization, thus simultaneously improving the overall energy density and efficiency (Figure S2). Note that the direct capacity contribution from KB is insignificant, less than 2 mAh g^{-1} at all testing rates (Figure S3). Interestingly, the KB-PP also enhances cell-to-cell capacity consistency for both NMC and LFP, especially at higher rates. Additionally, the same improvements

are not observed when KB is used instead as the conductive additive directly (Figure S4), nor are they observed with an anode-facing coating configuration, which persistently shorts and fails during charging (Figure S5). This indicates that the effect is largely specified to the cathode-side modulation, and confirms that it is not simply due to adding more conductive carbon to the electrode composite. Rather, the results suggest that the enhancement is more closely related to interfacial transport at the cathode-electrolyte boundary [29].

Cyclic voltammetry (CV) was employed on NMC and LFP half-cells to evaluate the effects of KB coating on redox behaviors of electrodes (Figure 2). The CV curves of a control stand-alone KB electrode (Figure S6) again confirm that KB itself contributes negligibly to the charge storage. Interestingly, both NMC and LFP electrodes with the KB layer exhibit broadened and less peak-dominated anodic responses with more gradual changes and reduced current responses, which seems counterintuitive given the improved Galvanostatic rate performance (Figure 2). This phenomenon can be rationalized by the formation of a diffuse layer within the nanoporous carbon coating. The thicker EDL compared to the case with plain PP separator requires prolonged charge redistribution, where during the CV's potential sweep, the pores cannot be fully equilibrated, leading to suppressed current response despite enhanced steady-state transport [21, 22, 25, 30–38].

2.2 | Deconvolution of the Interfacial Behavior Using DRT

The influence by the porous carbon coating on the interfacial charge-transfer kinetics for both NMC and LFP electrodes was studied via *operando* Galvano electrochemical impedance spectroscopy (GEIS) and the corresponding distribution of relaxation times (DRT) analysis. *Operando* GEIS measurements were performed from 1 MHz to 1 Hz, while conventional potentiostatic EIS (PEIS) was conducted from 1 MHz to 1 mHz at key state of charge (SoC). GEIS was limited to frequencies above 1 Hz to retain the efficacy of real-time spectral acquisition (Figure S7). Both half cells (using Li-metal as the counter/reference electrode) and symmetric cells were employed to separate the interfacial contributions from the lithium metal. Peak assignments within the DRT spectra were accomplished by comparative spectral analysis of the half-cell and symmetric-cell features of LFP and NMC at multiple SoC (Figures S8 and S9) together with the DRT spectrum of a Li||Li symmetric cell (Figure S10). All Nyquist impedance data are included in Figures S11 and S12.

Figure 3 compares the DRT spectra from GEIS to those from single-scan PEIS taken at 50% SoC for symmetric cells of both cathode materials. Peak component 1 (C1) corresponds to the resistance of the electrode/current collector/casing interfaces. Peak C2 represents the electron transfer of the charge transfer (CT) of one electrode, while C3 reflects the same event but with additional contributions from the counter electrode. Peak C4 is assigned to the cathode electrolyte interphase (CEI). Peak C5 is primarily contributed by the mass transfer of the charge transfer of one electrode, while peak C6 represents the counterpart of this process influenced by the counter electrode. Such mass-transfer events are interfacially coupled but spatially extended due to

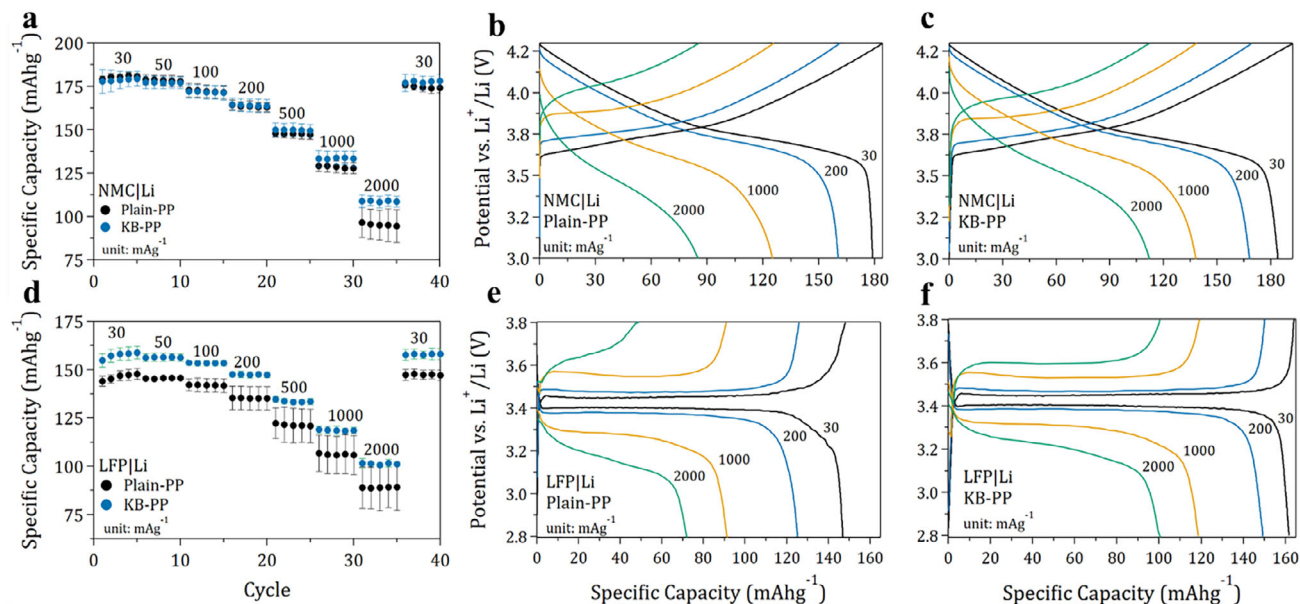


FIGURE 1 | Electrochemical performance of NMC and LFP with plain-PP and KB-PP. (a,d) Rate performance of (a) NMC||Li half cells and (d) LFP||Li half cells (three cells were tested for each cathode). (b,c) Representative GCD profiles of NMC at 30, 200, 1000, and 2000 mA g⁻¹ with (b) plain-PP and (c) KB-PP. (e,f) Representative LFP GCD profiles at 30, 200, 1000, and 2000 mA g⁻¹ with (e) plain-PP and (f) KB-PP.

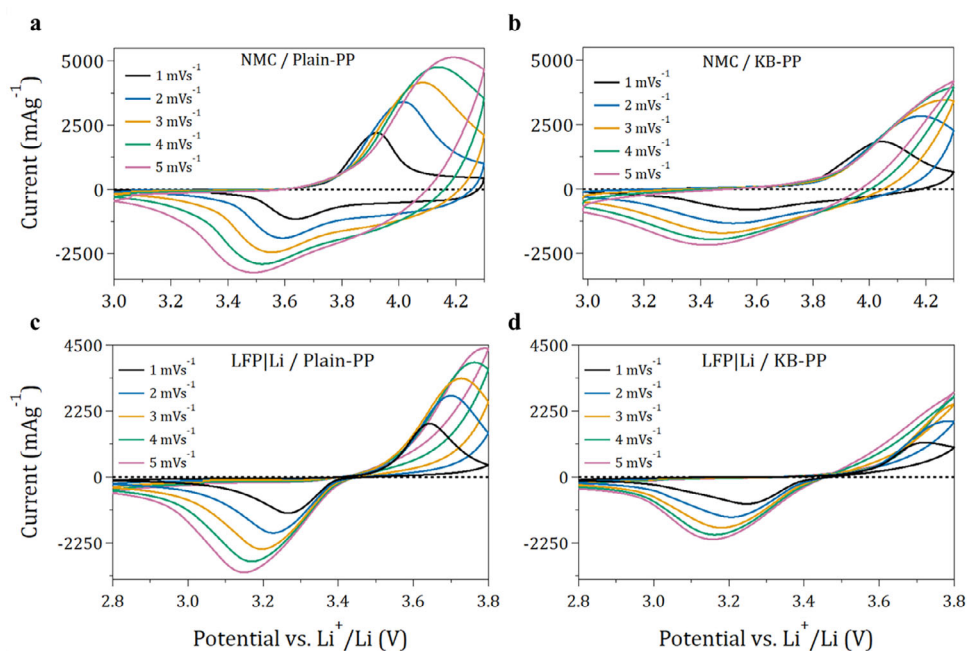


FIGURE 2 | CV profiles of NMC and LFP half-cells. (a,b) NMC||Li half-cells with (a) plain PP and (b) KB-PP at scan rates of 1, 2, 3, 4, and 5 mVs⁻¹. (c,d) LFP||Li half-cells with (c) plain PP and (d) KB-PP at scan rates of 1, 2, 3, 4, and 5 mVs⁻¹.

ion transport over finite distances. Peaks C7 and C8 represent the liquid-phase and solid-phase diffusion processes, respectively [39–48].

EDL plays a central role in regulating the CT kinetics at the cathode surface by modulating the local ion concentrations through both the Stern layer and the diffuse layer. These interfacial processes are commonly modeled using a Randles circuit, where $R = R_{CT}$ and $C = C_{EDL}$ represent the total charge-

transfer resistance and the total EDL capacitance, respectively [32–34]. According to the Gouy–Chapman–Stern model, the total capacitance follows $\frac{1}{C_{EDL}} = \frac{1}{C_S} + \frac{1}{C_D}$, where C_S and C_D are the Stern-layer and diffuse-layer capacitances. Because each capacitive component is associated with its own characteristic resistance, a continuous distribution of relaxation times for an extended RC network is expected to produce two distinct peaks in a DRT spectrum [18–23]. These peaks correspond to interfacial processes occurring at different frequencies: electron-transfer-

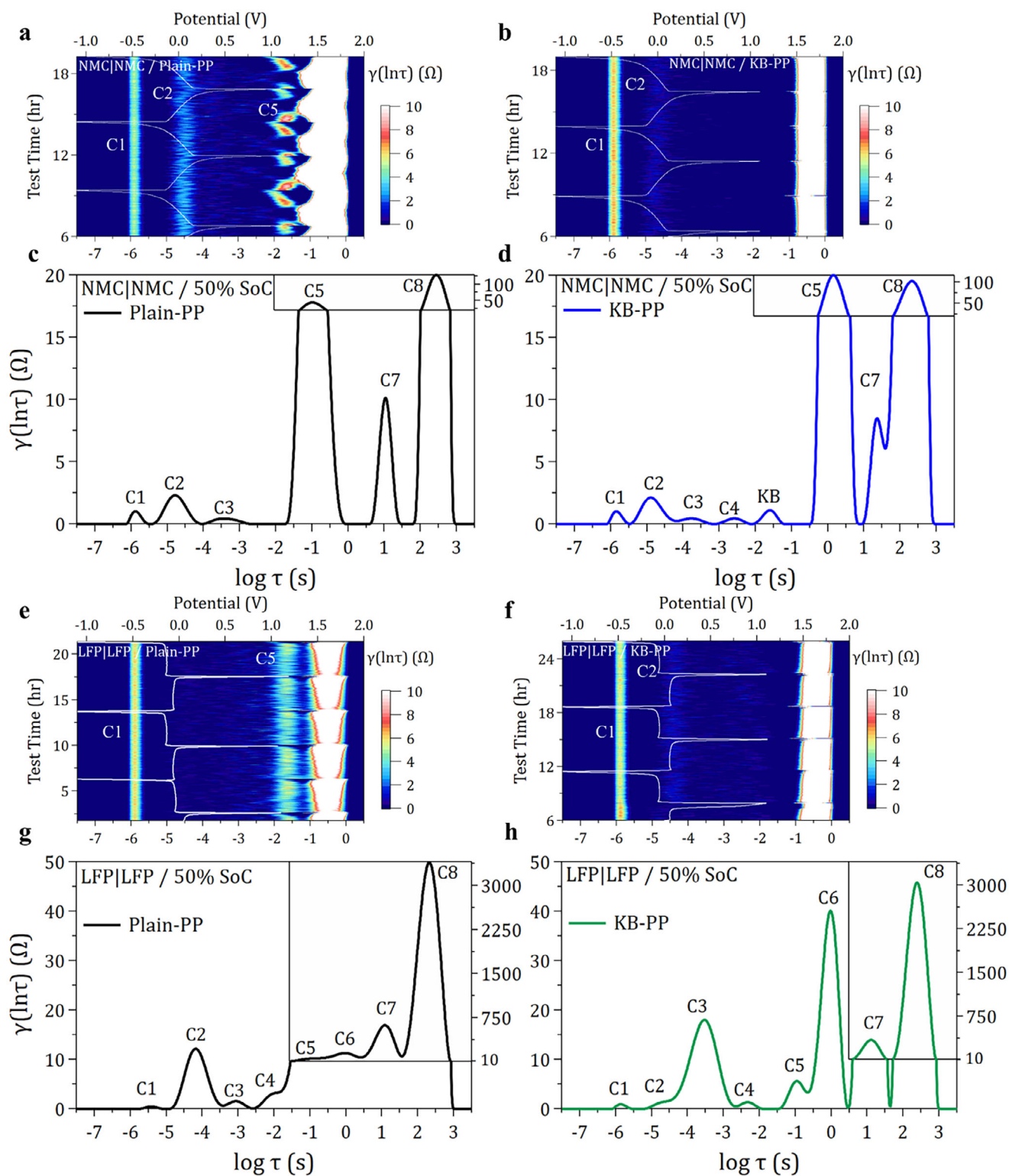


FIGURE 3 | DRT results of NMC and LFP symmetric cells. (a, b, e, f) *Operando* DRT (GEIS) from 1 MHz to 1 Hz of NMC||NMC symmetric cells with (a) Plain- and (b) KB-PP separators, and of LFP||LFP symmetric cells with (e) Plain-PP and (f) KB-PP separators. (c,d,g,h) The single-scan DRT (PEIS) from 1 MHz–1 mHz for NMC||NMC symmetric cells with (c) Plain-PP and (d) KB-PP separators at 50% SoC, and for LFP||LFP symmetric cells with (g) Plain-PP and (h) KB-PP separators at 50% SoC. The x-axes of the *operando* DRT are aligned with the axes of the single-scan DRT. Two separators are used so both the working and counter electrodes face a carbon layer.

dominated CT, C2/C3 peak, and mass-transfer-dominated CT within the diffuse layer, C5/C6.

The C1 peak shows a slight increase in magnitude for the KB-PP cells compared to the plain-PP cells, corresponding to the increased interfacial surface area caused by the KB-coating. For NMC, the C2 and C3 peaks in KB-PP cells are less resistive with slightly shorter τ values, suggesting faster electron transfer of CT. On the other hand, for LFP cells, KB-PP results in a more pronounced shift to shorter τ values for both peaks. The C2 peak decreases in magnitude, but the C3 peak increases in magnitude. The behavior disparity between NMC and LFP cells seems related to the inherently electronically insulating nature of LFP. The more prominent shift of τ values for LFP is considered due to a greater extent of improved electron transfer, mediated by the KB coating, in which NMC does not initially struggle. This interpretation is further supported by the LFP *operando* DRT where C2/C3 is only visible with the KB-PP cell (Figure 3e,f). A small peak observed in the NMC KB-PP system, labeled 'KB,' is tentatively assigned to an additional interface associated with the KB layer. This feature may reflect either the formation of a KB-related interfacial region at the separator side, behaving analogously to an interphase, or a CEI-related process in which the KB layer participates. This interpretation is supported by the absence of a C4 peak in NMC with Plain-PP. Moreover, the absence of this "KB" feature in the LFP system together with the more uniform and lower-intensity C4 peak, is consistent with the lower operating potential of LFP that suppresses electrolyte oxidation and thereby limits CEI formation. Once such an interphase is established, the lower oxidative driving force in LFP would also be expected to limit further electrolyte decomposition. Together, these observations suggest that the "KB" peak is associated with electrolyte-derived interfacial processes specific to the higher-voltage NMC system, rather than arising solely from an intrinsic property of the KB layer itself [49–52].

It is noteworthy that NMC shows no C6 peak at 50% SoC, despite its presence at other SoC (Figure S9), suggesting this event at the counter electrode is not rate-limiting at this SoC. With the KB layer, the mass-transfer resistance peak C5 is decreased in LFP but is slightly increased in NMC, which aligns with NMC's slightly poorer performance at low rates with KB-PP (Figure 1a). Both electrode materials show an increase in the corresponding relaxation time τ for C5. This seems counterintuitive since a decrease in resistance should yield a shorter τ , following the relationship of $\tau = R \cdot C$. Therefore, this behavior can only be attributed to an overwhelming increase in capacitance, which is consistent with the CV results and supports the interpretation that the lower CV current response stems from insufficient charging time needed for a thicker and more charge-dense diffuse layer.

Peak C7 shows that the KB layer slightly decreases the liquid-phase ion diffusion resistance. For solid-phase ion diffusion (peak C8), LFP shows substantially higher inherent resistance than NMC, consistent with its distinct two-phase redox mechanism compared to the more gradual solid-solution mechanism for NMC. Interestingly, the KB-layer slightly decreases the solid-phase diffusion resistance for both electrodes, which may be due to a more uniform and consistent gradient of Li-ions at the cathode interface.

2.3 | Li-Ion Transport in Simulated Pore Model

To elucidate how the nanoporous carbon influences Li-ion transport, we developed a simplified pore-scale model based on the size-modified Poisson–Nernst–Planck equations, capturing the dominant EDL effects within the pore and quantifying the resulting Li-ion flux. Simulations across a range of pore sizes were computed to evaluate how pore size modulates the Li-ion transport behavior [27, 28]. Figure 4a illustrates the schematic of the pore model, where the nanopore, B, lies between two reservoirs, A and C. The geometry is modeled as a symmetric, 2D slit of varying widths, where w is the pore width, $H = 8w$ is the reservoir height, and the length of each reservoir and the nanopore is fixed at $L = 50$ nm. A Li^+ concentration gradient is applied from right to left between reservoirs, which serves as the primary driving force for Li-ion transport. To approximate a cathode operation, a fixed potential of 0.414 V with respect to the left reservoir outlet is applied at the nanopore wall (region D). More details can be found in the Methods.

Figure 4b shows the total flux of Li-ions along the centerline of the pore ($y = \frac{H}{2}$) across a distance of $3L$ with varying pore widths, w . The local flux increases as the pore narrows but drops sharply at $w = 2$ nm. This is explained by the concentration gradients of Li^+ (Figure 4c) and PF_6^- (Figure S13) along the centerline of the pore, both of which get steeper with decreasing pore width. The same trend is observed for the total flux across the EDL (Figure 4d), indicating that the flux is more localized toward the center of the pore as the pore size decreases. This is further explained by the respective ion concentrations (Figure 4e) and concentration gradients (Figures S14 and S15) in the EDL. As the pore size decreases, the Stern-layer in the EDL and adsorbed anions occupy an increasingly larger volume of the nanopore, causing the diffuse layers from opposing walls to overlap. This suggests that the EDL structure, particularly the degree of diffuse-layer overlap, directly influences the Li-ion flux in the nanopore.

To further elucidate the influence of the EDL, the total flux was deconvoluted into the diffusion (Figure 4f) and migration (Figure 4g) contributions. The diffusive flux follows the same trend as the total flux but shows a unique bump immediately following the initial increase. This is explained by the formation of the Li^+ concentration gradient near the wall of the pore (Figure S14). In contrast, the migration flux shows a maximum centerline flux with a pore width of 12 nm, and the flux at the centerline of the 2 nm pore is substantially worse. Since the migration flux shows a substantially lower flux at 2 nm and the 3 nm pore has the highest total centerline flux, it can be inferred that over confinement of the EDL impedes Li-ion migration. This notion is supported in part by Figure 4e, where it is seen that at $w = 2$ nm the potential field is still significantly present at the center of the pore and, at both 2 and 3 nm, the Li^+ concentration remains below bulk and is overwhelmed by a consistently larger PF_6^- concentration. Furthermore, as shown in Figure 4c, the 2 and 3 nm pores exhibit competing concentration gradients, which induces a subsequent competing potential force (Figure S16) responsible for the observed impact on migration flux. This suggests that overly confined pores induce overlapping potential fields and EDLs that negatively impact the flux of Li-ions.

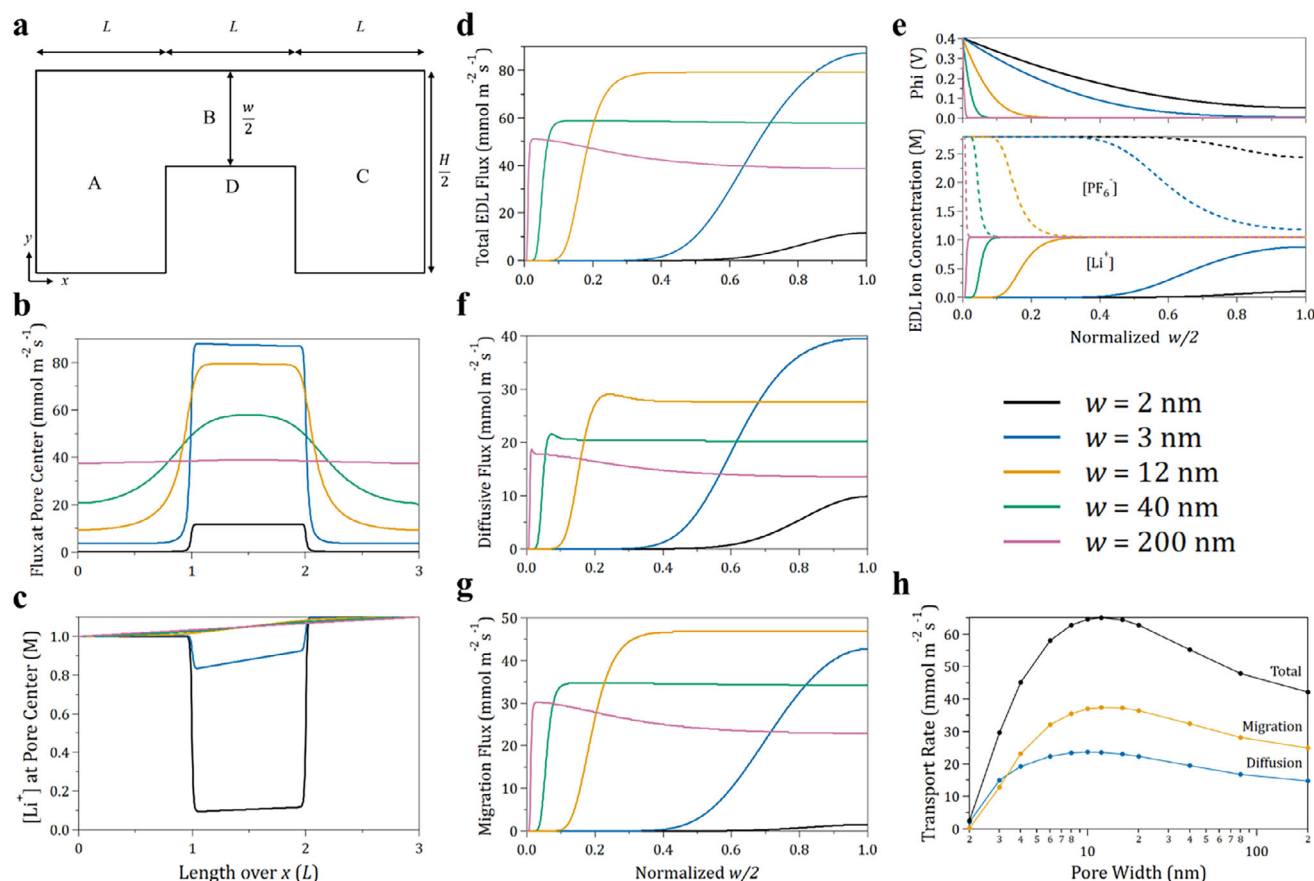


FIGURE 4 | Simulations of ion transport through nanopores. (a) Schematic of the pore model. (b) Total Li^+ flux along the center of the pore. (c) Li^+ concentration along the center of the pore. (d) Total Li^+ flux across the EDL from the pore wall to the pore center. (e) Potential in volts (top) and Li^+ and PF_6^- concentration (bottom) across the EDL from the pore wall to the pore center. (f) Diffusion-specific contribution to the total EDL flux. (g) Migration-specific contribution to the total EDL flux. (for d, e, f, and g, 1.0 marks the centerline of the pore). (h), Integrated Li^+ transport rate derived from areas under the curves in (d), (f), and (g), and additional pore sizes.

To better quantify the impact of this EDL behavior, the areas under the total, diffusive, and migration flux curves across the EDL were calculated and normalized by their respective pore width (Figure 4h). This analysis reveals a maximum transport of Li-ions with a pore size of $w = 12$ nm, which is a result of the interplay of two competing effects. The first effect is the suppression of Li-ion occupancy and the reduced effective cross-section of the nanopore due to overlapping EDLs, which hinders transport [53]. The second effect is an increase in integrated flux resulting from a reduced diffusion resistance. Because the ratio between H and w is fixed and any frictional forces at the walls are neglected, the geometrical setup yields a reduced diffusion resistance (Figure S17) [54]. Since the migration and diffusion fluxes are closely related via the no-flux boundary condition for PF_6^- (Equation (8) in the Methods section), this reduced resistance accounts for the increase in transport observed as pore size decreases from 200 nm to around approximately 12 nm. This feature is specific to the model geometry and should therefore be considered when interpreting the resulting maximum in transport. For comparison, the equivalent curve for the total flux in Figure 4h for a straight pore with $H = w$ is shown in Figure S18. In that case, the transport rate increases with pore size between 2 and 12 nm and then saturates at larger pore sizes.

Additional simulations of a non-polarizable porous coating, in which no potential is applied at the pore walls, as well as alternative geometries show a continued increase in flux with a decreasing pore size (Figures S19–S25). This trend arises from the same geometrical reduction in diffusion resistance described above. We further note that the flux may stop increasing or even begin to decrease as the pore sizes become closer to the ion sizes due to additional unrepresented physical processes (e.g., ion-ion and ion-solvent frictions) [55–58]. However, most of the simulated pore sizes are order-of-magnitude larger than the ion and solvent sizes (Table S2) and thus the observed phenomena remain useful for mechanistic understanding and material design.

Although the geometry is simplified, the model captures the behavior of the EDL and helps explain its mechanistic influence on the Li-ion transport through nanopores. These results suggest that as confined EDLs approach overlap within the pore, the local increase in concentration gradient of Li-ions at the center of the pore acts analogously to a converging nozzle shaped by the potential field across the EDL accompanied by a reduction in area available for Li^+ transport in the nanopore. This indicates that the pore size may be optimized by expanding the pore size of the KB coating, thereby reducing excessive Li^+ exclusion by the EDL.

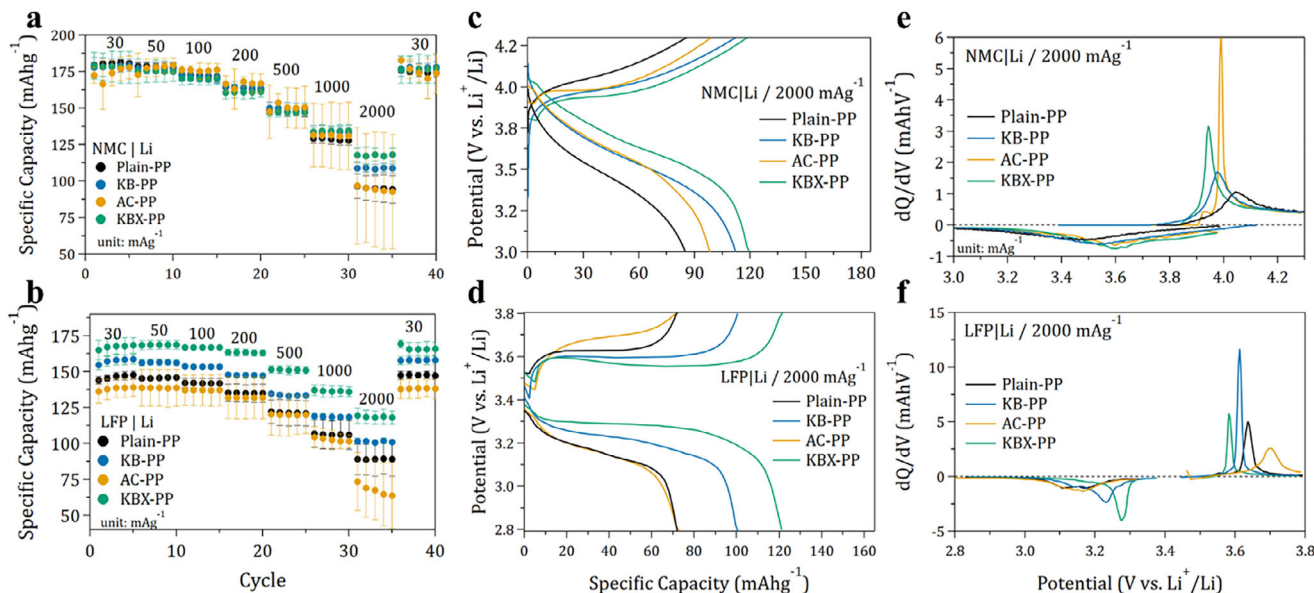


FIGURE 5 | Electrochemical performance of NMC and LFP with separators of different coating conditions. (a,b) Averaged GCD rate performance of (a) NMC and (b) LFP half-cells comparing plain-PP, KB-coated, AC-coated, and KBX-coated PP separators. (c,d) GCD potential profiles of (c) NMC and (d) LFP from the closest to average cells at 2000 mA g⁻¹. (e,f) Corresponding dQ/dV profiles for (e) NMC and (f) LFP with a plain, KB-, AC-, and KBX-coating.

2.4 | Simulation-Guided Experimental Design

To experimentally evaluate the simulation predictions regarding the effects of over-confined versus less-confined pores, two additional carbon coatings for the PP separator were prepared. High surface area activated carbon (AC-PP) was selected to represent the over-confined pore size condition, where a large portion of its pores are smaller than 2 nm, compared to KB with pores sized between 3 and 4 nm (Figures S26 and S27). Super C65 was used as a low-porosity carbon control with a morphology similar to that of KB (Figure S28). To obtain a pore size distribution closer to the simulated maximum, the KB pore size was expanded using a CO₂-loaded inlet-controlled calcination at 900°C, which results in an oxidized form of KB, referred to as KBX, while KBX-2 refers to KB subjected to two consecutive calcination treatments. Surface area and pore-size distribution analyses reveal that both KBX variants contain a much greater volume of pores larger than 4 nm of the pristine KB. Specifically, KBX has pores sized ~5–7 nm, while KBX-2 also shows an additional increase in pores larger than 20 nm (Figure S29). Although this method did not produce pores centered precisely at 12 nm pores, this allows the comparison of KB-derived carbons with larger pores than pristine KB, approaching the simulated optimal pore-size regime from both smaller and large pore sides.

The AC coating did not out-perform the initial KB coating for cathodes (Figure 5a,b) whereas the KBX and KBX-2 produced the highest improvement to rate performance among the carbon coatings (Figure S30). KBX-2 shows slightly higher average capacity, but with a minor increase in cell-to-cell variation compared to KBX, which we attribute to the broader pore size distribution generated by repeated calcination. With the KBX coating on the separator, NMC and LFP reached capacities of 121 and 125 mAhg⁻¹ at 2 Ag⁻¹ (~12 C), respectively (Table S1 and Figure S31)

[59–64]. Interestingly, the AC-PP produced a large variation of capacity values between parallel testing cells in both systems, especially NMC. With AC-PP, the average capacity of NMC showed only slight improvement at low rates, but decreased at higher rates, while LFP performed consistently worse than the KB coating at all rates. The poor performance of AC-PP aligns with the expectation from the simulation results due to the smaller pore size. In addition, AC particles are bulk-sized, where their nanopores are tortuous for Li-ions to meander through, and such thick, randomly packed EDLs add impedance for cell operation.

To evaluate whether improved wettability by the electrolyte contributes to the enhanced electrochemical performance, we conducted contact-angle measurements of the non-coated and carbon-coated separators (Figure S32). The results confirm substantially improved wettability after carbon coating, with the contact angle decreasing from ~32° for Plain-PP to ~0° for KB-, AC-, and KBX-PP. However, improved wettability alone does not fully account for the observed rate enhancement. To decouple wettability effects from pore-transport effects, we prepared a low-porosity carbon-coated separator using Super C65 (SC-PP), which has a morphology similar to that of KB but a lower pore volume (Figures S28 and S29). Although SC-PP also exhibits improved wettability (Figure S32), it performs worse than the plain separator at most current rates, with only a minor improvement observed at 2 A g⁻¹ in LFP cells (Figure S33). In contrast, KBX-PP provides substantially better rate performance. These results indicate that improved wettability is beneficial but is not the dominant factor governing the performance enhancement. Instead, the improved rate capability is more closely associated with ion transport through the nanoporous carbon coating. Moreover, the KBX-PP has only a limited impact on the cycle life compared to the plain separator (Figure S34). After 200 cycles at 2 C, the capacity retention remains unaffected for both NMC and LFP, with differences within ≤ 0.3%. These results

support that the carbon coating primarily improves interfacial transport processes, but does not affect bulk cathode properties, particularly the cycling stability.

On the other hand, the voltage profiles (Figure 5c,d) and dQ/dV (Figure 5e,f) reveal that AC-PP significantly increases overpotential in both systems, which also correlates with the specific impact on migration flux observed in the simulation data at over-confined pore sizes. When transport is restricted, a larger overpotential is necessary to drive sufficient transport of Li-ions. In contrast, the KBX markedly reduces the overpotential in both systems, even compared to the initial KB-PP. This reduced polarization, especially under high current rates, is consistent with a more uniform Li-ion distribution at the cathode interface. Together with the DRT results, this observation further supports that the performance improvement is associated with interfacial mass transport rather than increased electronic conductivity of the cathode. This effect is expected to become most pronounced under high-rate conditions, where non-uniform ion supply and concentration gradients become more significant.

3 | Conclusions

This study demonstrates that coating a Ketjenblack nanoporous carbon layer on the cathode-facing side of a polypropylene separator enhances the rate performances of both NMC and LFP in LIBs. The experimental results suggest that the nanoporous carbon coating increases Li^+ concentration in the diffuse layer of the EDL at the cathode interface. The simulation results cannot directly explain the experimentally observed improvement to the rate performance but predict a non-monotonic correlation between Li-ion transport rate and carbon pore size with a local maximum around 12 nm. The combined interpretation suggests that the porous conductive carbon layer provides nanoconfined electrolyte-filled domains that promote more homogeneous Li-ion distribution and facilitate interfacial ion transport. Specifically, as the EDLs are confined within the nanopores, the modified potential field acts analogously to a converging nozzle to accelerate the transport of Li-ions and enhances both (de)lithiation processes by promoting a more uniform and consistent gradient of Li-ions, thus reducing the effective cross-section for Li^+ transport. As a result, polarization is reduced at high currents, which enables improved utilization of the active mass, especially for more kinetic/diffusion-limited materials such as LFP. By expanding the carbon pore size, we observe 121 mAhg^{-1} for NMC and 125 mAhg^{-1} for LFP at a rate of 2 Ag^{-1} ($\sim 12^\circ\text{C}$). Moreover, this work implies that the effect of improving rate capability by coating porous carbon on the separator may be independent of the electrode material by directly enhancing Li-ion transport rate at the electrode/electrolyte interface. Therefore, this separator coating strategy may show benefits beyond Li-ion systems.

4 | Experimental

4.1 | Fabrication of Carbon-Coated Separators

The KB-PP separator was prepared using a slurry method consisting of an 8:2 mass ratio of KB carbon (Ketjenblack EC600

JD, MSE Supplies LLC) to polyvinylidene fluoride (PVdF, MTI Corporation, 600 kDa, >99.5%) in N-methyl-2-pyrrolidone (NMP, TCI America, >99%). KB (160 mg) was hand ground using an agate mortar and pestle and added to a 25 mL Erlenmeyer flask with a polytetrafluoroethylene (PTFE) coated stir bar. Next, 400 μL of a PVdF in NMP solution (50 mg/mL) was added to the flask followed by additional NMP (5 mL). The flask was sealed with a rubber septum and left stirring for a minimum of 24 h. An appropriate amount of slurry was then placed onto a strip of Celgard 2500, spread using a 100 μm doctor blade and dried under vacuum for 12 h at 80°C . The dried and coated separators were then punched into 19-, 15-, and 12 mm diameter disks. The average mass loading and thickness of the 19 mm carbon coatings was 0.73 mg and 9 μm and was further measured by cross-sectional scanning electron microscopy as 10 μm (Figure S1). The AC-PP separator was prepared with High Surface Area Activated Carbon (MTI Corporation) using the same conditions as KB-PP, with the only difference being the amount of additional NMP (1.5 mL). The KBX separator was prepared using the same conditions as KB-PP.

4.2 | Fabrication of Cathode Electrodes

$\text{LiNi}_{0.6}\text{Mn}_{0.2}\text{Co}_{0.2}\text{O}_2$ (NMC622, MTI Corporation) and LiFePO_4 (LFP, MTI Corporation) cathode slurries were prepared using an 8:1:1 mass ratio of active material, PVdF, and carbon black (Super C65, MSE Supplies LLC). For each, active material powder (16 g) and Super C65 (2 g) were hand ground using an agate mortar and pestle and combined with a 0.1 g/mL PVdF in NMP solution (20 mL). Additional NMP (20 mL) was added, and the collection was vacuum mixed using an MTI MSK-SFM-16 vacuum slurry mixer until homogenous. The slurry was coated onto carbon-coated aluminum current collector (Double-Sided Conductive Carbon Coated Aluminum Foil, MSE Supplies LLC) using a 150 μm doctor blade and dried under vacuum for 12 h at 80°C . The electrodes were punched into 12 mm disks with active loadings between 3.12 to 3.85 mg for NMC and 2.78 to 3.35 mg for LFP. The cell-specific active mass loadings were measured at the time of cell assembly.

4.3 | Coin Cell Assembly

CR-2032 coin cells, assembled in an Ar-filled glove box, were used for all electrochemical tests. Construction of all NMC||Li and LFP||Li half cells used 12 mm (diameter) cathodes, 70 μL electrolyte (1 M LiPF_6 in 50/50 by wt. ethylene carbonate/diethyl carbonate, Gotion), plain or coated polypropylene separator (Celgard 2500); 15 mm (diameter) lithium metal counter electrode; 316 stainless steel (SS) current collector; 316 SS disc springs.

NMC||NMC and LFP||LFP symmetric cells were made by constant current charging, discharging, and charging of two cathodes to the respective state of charge (SoC) being tested. The cells were then disassembled in an Ar-filled glove box and conjoined to form the corresponding SoC symmetric cell. For *operando* electrochemical impedance spectroscopy (EIS), the symmetric cells were constructed with each electrode at 50% SoC for consistency in electrode conditioning.

4.4 | Characterizing Carbon Pores

Surface area analysis and pore size distribution (PSD) were conducted using a Tristar II Brunauer–Emmet–Teller (BET) analyzer from Micromeritics. PSD was calculated from the N_2 sorption isotherm data using Barrett–Joyner–Halenda (BJH) theory with Faas correction [65].

4.5 | Electrochemical Characterization

All Galvanostatic charge-discharge (GCD) cycling and Coulombic efficiency (CE) testing were performed on a Landt CT2001A in a Landt LBI-300HT battery test chamber at 30°C.

Cyclic voltammetry (CV) and EIS experiments were performed using a BioLogic VMP3 potentiostat. The diffusion coefficient obtained by CV was calculated using the Randles–Sevcik Equation [33].

$$i_p = 2.69 \times 10^5 n^{\frac{3}{2}} A D^{\frac{1}{2}} C v^{\frac{1}{2}} \quad (1)$$

where n is number of electrons transferred, A is electrode surface area, C is concentration of redox species, v is the scan rate, and D is diffusivity constant.

The distribution of relaxation times (DRT) (using the Infinite Voigt Circuit method) was employed using the DRTtools freeware developed by T. H. Wan et al. to deconvolute the impedance data [26]. The parameters of the model included fitting with inductance, Gaussian discretization, and 2nd order-regularization. The expression for fitting the EIS data is shown in Equation (2),

$$Z_{\text{DRT}} = R_{\infty} + i\omega L + \int_{-\infty}^{\infty} \frac{\gamma(\tau)}{1 + i\omega\tau} d\tau \quad (2)$$

where Z_{DRT} is the impedance, R_{∞} is the Ohmic resistance, $\gamma(\tau)$ is the distribution function of relaxation time τ , and L is the inductance contribution. And τ is calculated from the resistance R , and capacitance C , components from each Randles element in the infinite Voigt model as $\tau = R \cdot C$.

Operando DRT was collected by performing 1 MHz–1 Hz EIS scans every 65 s throughout Galvanostatic cycling. To obtain single-scan DRT, cells underwent three conditioning Galvanostatic cycles, then charged to a specific SoC and rested for 24 h at 30°C before collecting impedance data from 1 MHz–1 mHz. Importantly, the reported SoC of 10%, 50%, and 100%, correspond to 3.6, 3.8, and 4.3 V, respectively, for NMC622. And 0.044 mAhg⁻¹, 0.22 mAhg⁻¹, and 3.8 V, respectively, for LFP.

4.6 | Computational Simulations

The pore model structure is represented by a simplified three-channel system (Figure 4a) with a slit-pore between two reservoirs. We further assume a symmetry along the center of the pore and therefore only consider the lower half. The domain is occupied with 1 M of solvated Li⁺ and PF₆⁻ in a model carbonate solvent. The ion transport in the electrolyte is assumed to be governed by the size-modified Poisson–Nernst–Planck equations,

$$\frac{\partial c_i}{\partial t} = \nabla \cdot (M_i c_i \nabla \mu_i), \quad (3)$$

$$\nabla^2 \phi = -\frac{1}{\varepsilon} \sum_{i=1}^2 z_i e c_i \quad (4)$$

where $i = 1, 2 = \text{Li}^+, \text{PF}_6^-$, c_i is the concentration of the i^{th} ion, t is time, $M_i = \frac{D_i}{k_B T}$ is the ion mobility where D_i is the diffusion coefficient, k_B is the Boltzmann constant, T is the temperature, ε is the permittivity, ϕ is the electric potential, z_i is the valence, e is the elementary charge, μ_i is the electrochemical potential, and μ_i that accounts for the ion size effect is given by

$$\mu_i = k_B T \ln(c_i a_i^3) - \frac{k_B T a_i^3}{a_0^3} \ln(1 - \sum_j c_j a_j^3) + z_i e \phi \quad (5)$$

where a_i is the excluded size of the i^{th} species and a_0 is the excluded size of the solvent molecule [27, 28].

Equations (3)–(5) represent the governing equations for multi-species ion transport in the electrolyte. We impose a concentration gradient of the Li⁺ ions as the driving force from right to left between the reservoirs and specify a zero-flux condition for PF₆⁻ which yields

$$c_{\text{Li}^+} = c_{\text{PF}_6^-} = 1 \text{ M}, \text{ at } x = 0 \quad (6)$$

$$c_{\text{Li}^+} = 1.1 \text{ M}, \text{ at } x = 3L \quad (7)$$

$$\mathbf{n} \cdot \mathbf{J}_{\text{PF}_6^-} = 0, \text{ at } x = 3L \quad (8)$$

where $\mathbf{J}_i = M_i c_i \nabla \mu_i$ is a 2D vector representing the flux of the i^{th} ion. The remaining boundaries are either solid surfaces (where ions cannot penetrate) or symmetric boundaries (where ion fluxes are zero in the normal direction). Therefore, we apply zero normal fluxes for those boundaries, which yields

$$\mathbf{n} \cdot \mathbf{J}_i = 0. \quad (9)$$

For the electric potential, we specify a reference potential of 0 V at the outlet and a zero potential gradient at the inlet,

$$\phi = 0, \text{ at } x = 0 \quad (10)$$

$$\mathbf{n} \cdot \nabla \phi = 0, \text{ at } x = 3L. \quad (11)$$

For the remaining boundaries, we assume zero normal gradient of electrical potentials, which yields

$$\mathbf{n} \cdot \nabla \phi = 0. \quad (12)$$

Alternatively, for the pore wall of carbon coating, we apply a fixed electrical potential,

$$\phi = 0.414 \text{ V}, \text{ at } y = \frac{H}{2} - \frac{w}{2}, \text{ and } L \leq x \leq 2L. \quad (13)$$

We solve the governing equations for the boundary conditions numerically using a backward Euler finite volume method. The numerical algorithm is implemented using the Julia language,

built upon an open-source package named VoronoiFVM.jl [66, 67]. The package adopts Voronoi meshes for spatial discretization and uses a two-point flux approximation for computing the ionic fluxes between neighboring numerical cells. The resulting non-linear system of equations is solved through a Newton–Raphson algorithm where the Jacobian is computed using automatic differentiation [68]. A sufficient grid resolution in the nanopore was confirmed by a convergence study (Figure S35).

We performed simulations for coated and uncoated conditions for 12 pore sizes ($w = 2, 3, 4, 6, 8, 10, 12, 16, 20, 40, 80$, and 200 nm). Each simulation is run until the ion transport reaches steady state. The domain dimensions and such parameters as pore sizes, reservoir height, pore length, ion diffusivities, solvent size, and ionic radii are provided in Table S2 [69–77]. And the linear solver, non-linear solver, and time stepping tolerance values are provided in Table S3.

The steady-state solutions are used to compute the transport rates in Figure 4h according to

$$\text{TransportRate} = \frac{2}{w} \int_{\frac{H}{2} - \frac{w}{2}}^{\frac{H}{2}} J_{x, \text{Li}^+} \left(x = \frac{3L}{2}, y \right) dy \quad (14)$$

where J_{x, Li^+} is the total flux as defined above or the diffusion, migration or size-term specific contribution, respectively. We verify by a set of simulations where $\frac{w}{H}$ ranges from 1 to $\frac{1}{12}$ in Figure S36, that for $\frac{w}{H} \leq \frac{1}{8}$, there is minimal influence on the flux, so for the final simulations we set a constant reservoir height of $H = 8w$.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adma73871-sup-0001-SupMat.docx.